

Robotics Project Report

Project Title: Robotic Arm Design and Development

Team Members:

1. K.AZAAN AHAMED (23BEC0362)
2. HIMASUNDAR NAIDU (23BEC0467)
3. TALASU SANJAY (23BEC0301)

Objective:

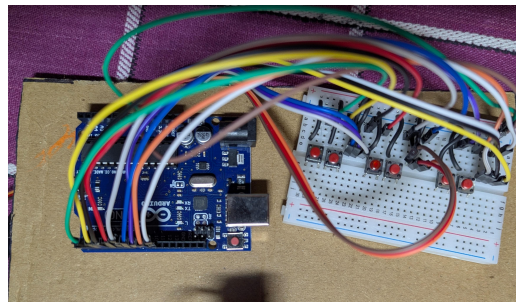
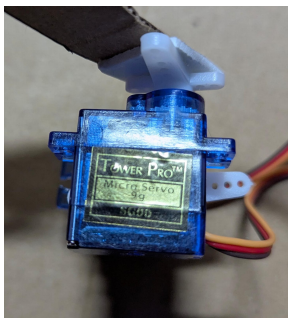
To design and construct a functional robotic arm capable of picking up and manipulating objects using the Arduino platform and servo motors.

Introduction:

This project involves the application of engineering principles and the use of Arduino microcontrollers to create a robotic arm with multiple degrees of freedom [3D] . The project emphasizes iterative design and hands-on prototyping, culminating in a functional robotic system tailored for educational demonstrations.

Materials Used:

- 1.Arduino board
- 2.Servo motors (3)
- 3.Breadboard and wires
4. Cardboard
- 5.Double-sided tape



Methodology:

1. Design Phase

Brainstormed the arm's functionality and identified key requirements.
Created a hand-drawn sketch of the arm layout, indicating motor placement and gripper design.

2. Circuit Design and Testing

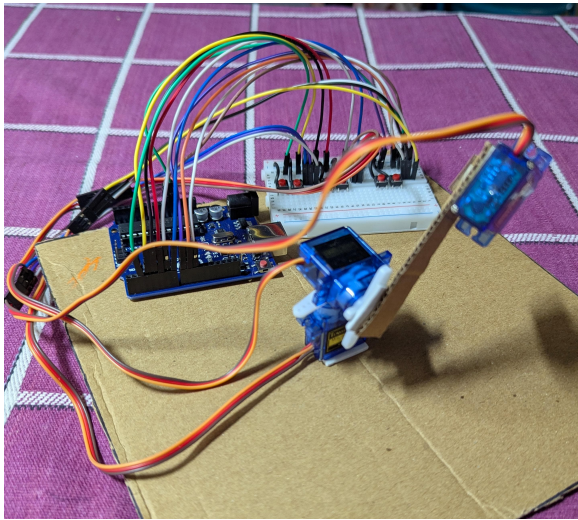
Built and tested the circuit to control servo motors using buttons.
Ensured the motors responded accurately to inputs using example Arduino code and customized the code as needed.

3. Assembly of Robotic Arm

Constructed the arm using sturdy materials and secured motors to the base and arm parts. Implemented wire management strategies to prevent tangling during operation.

4. Testing and Iteration

Conducted tests to assess the range of motion, stability, and gripper functionality. Modified the design to address issues such as limited motion or inadequate grip.



Results:

The robotic arm successfully performed picking up small objects and placing them accurately. It withstood the maximum weight of 100grams.

Challenges Faced:

It was little difficult in wiring between the breadboard and the Arduino board ,but it resolved by our teammates.

Conclusion:

The project achieved its objectives by creating a functional robotic arm capable of picking up and manipulating objects. The iterative engineering design process allowed the team to address challenges and improve the arm's performance. Future improvements could include using stronger motors or adding sensors for automation.

References:

Science Buddies Engineering Guide